

Drug Clustering in a Private Hospital of Chiang Mai Province with K-Means Clustering Technique

Parinthewat Pintakiew¹ ; Wimalin Laosiritaworn¹



Parinthewat Pintakiew

¹ Department of Industrial Engineering, Chiang Mai University, Chiang Mai, Thailand.

* Address Correspondence to author:
Wimalin Laosiritaworn
Department of Industrial Engineering,
Chiang Mai University,
Chiang Mai, 50200, Thailand.
email: wimalin@hotmail.com

Abstract

OBJECTIVES: The study was to establish a methodology for clustering medicines into distinct groups, hence facilitating effective management within each category.

MATERIALS AND METHODS: This study employed the k-means clustering technique to examine the drug inventory data of a hospital. This research gathered data from a private hospital located in Chiang Mai province, Thailand. The dataset consisted of 1,501 unique items that were monitored over a period of 24 months. Each item was described by parameters such as importance, annual drug expenditure (ADE), variance, and mobility. The clinical importance of the medicine was determined by Vital, Essential, Desire Analysis (VED analysis), which involved input from healthcare professionals such as physicians, pharmacists, and inventory personnel. Subsequently, the chosen medication category was subjected to k-means clustering using the RapidMiner software, which also contributed to data preparation, transformations, and the identification of appropriate parameters for the clustering procedure.

RESULTS: The VED analysis shows the result of clinical importance grouping which are vital (213 items), essential (588 items), and desirable (700 items). Only the essential group is focused on and processed by k-means clustering analysis, due to higher complexity and contribution to various patient outcomes. The clustering analysis yielded six discrete clusters of drugs, each exhibiting distinct characteristics, and management strategies have been recommended for each cluster. Some clusters can be handled using simple methods or existing inventory management policy, while others need the use of statistical forecasting tools and/or machine learning techniques to precisely predict demand.

CONCLUSION: This study provides a baseline and framework for future studies and the management of drug inventory by using mathematical methods and models. The use of VED analysis also contributes significant data clarification in terms of clinical importance to the hospital. Integrating k-means clustering is one of the current machine learning technologies for inventory management, which is needed to improve the efficiency of patient care and to ensure more effective management of drugs. As a result, by introducing a new clustering technique, hospital staff members in charge of managing the drug inventory gain new insight and guidelines into inventory policy, which they can then apply as circumstances change and implement further.

Keywords: hospital inventory, k-means clustering, VED analysis, machine learning

An important objective of hospital administration is to provide superior healthcare services while ensuring high-quality standards are met. An essential component in achieving this goal is maintaining a complete and sufficient inventory of pharmaceuticals, as this is fundamental to facilitating the functionality of hospital personnel.

Typically, inventory management within healthcare institutions encompasses the dispersion of products to multiple units throughout the hospital. Moreover,

Received: February 23, 2024
Revision received: June 28, 2024
Accepted after revision: August 26, 2024
BKK Med J 2024;20(2): 98-108.
DOI: 10.31524/bkkmedj.2024.21.008
www.bangkokmedjournal.com

these products are often stockpiled in substantial quantities to ensure the prompt fulfillment of patient needs and to guard against product insufficiencies.¹

The issue of pharmaceutical product diversity, and the principle of economies of scale, constitutes a central concern within the realm of hospital inventory management. The pertinent stakeholders in this context encompass medical practitioners, pharmacists, and the procurement department. Medical practitioners frequently need the ability to select and employ a wide range of medications to ensure the provision of optimal patient care under varying clinical circumstances. They may also be subject to influence from pharmaceutical manufacturers or emerging treatment protocols, potentially leading to the inclusion of new pharmaceuticals in the hospital's medication catalog. On the other hand, pharmacists are tasked with the reduction of the hospital's medication inventory to curtail associated carrying costs and management expenditure, including procurement. Meanwhile, the procurement department seeks to economize by acquiring cost-effective medications for the hospital.²

The research done by Abu Zwaïda, Pham³ finds that the management of hospital inventory has issues that might result in drug shortages. Several factors contribute to these concerns, including an erroneous evaluation of demand, an ineffective inventory management system, difficulties with purchasing and delivering items, and insufficient technology and information systems. These impediments may lead to an extended waiting period for patients to get necessary drugs, thus compromising the quality of their treatment. The study highlights the significance of customized inventory management systems to maintain a consistent provision of healthcare services and minimize occurrences of pharmaceutical shortages.

This research utilizes data from a private hospital in the Chiang Mai province, Thailand, where several significant challenges have been observed in the domain of pharmaceutical data management. These challenges cover a spectrum of issues, including medication shortages, incomplete and uneven pharmaceutical inventory management practices, as well as an inherent inability to effectively prioritize and optimize medication management. An added layer of complexity is introduced by the presence of drugs that reach their expiration dates or undergo devaluation, primarily due to the absence of a systematic framework for the categorization and prioritization of medications. However, this study cannot provide information regarding patient volume, medication distribution, and pharmacy staff, as this data is confidential within the hospital.

In this hospital environment, the classification and ranking of medication importance heavily relies on the experiential judgment and discretion of healthcare personnel, such as physicians and pharmacists, who play integral roles in the decision-making process regarding pharmaceutical usage. It is noteworthy that this hospital maintains an extensive inventory, comprising in excess of 1,501 distinct pharmaceutical

items, rendering the task of effectively categorizing and forecasting the entirety of the medication inventory an onerous and potentially futile undertaking. Compounding the complexity of pharmaceutical management in this context is the notable divergence in therapeutic properties and financial value associated with each pharmaceutical item present in the hospital's repository. Each medication's unique attributes necessitate tailored and distinct approaches to management, thereby obfuscating any attempt to implement a unified decision-making model for medication management.

This research, therefore, seeks to address these challenges by proposing the systematic clustering of medications using k-means algorithm with comparable attributes into well-defined categories. Subsequently, the research aims to identify and prioritize those categories that hold particular significance for the hospital, thereby offering a more structured and efficient framework for medication management and optimization within the hospital's operations.

Categorization approach in pharmaceutical research

Numerous research studies have employed a categorization approach in pharmaceutical research that involves the utilization of Annual Drug Expenditure (ADE) data and employs Always, Better, Control Analysis (ABC Analysis) as tools. Analysts are required to determine the proportion of ADE to represent the value and cost of each type of medication, utilizing historical data. In addition to assessing the importance of individual medications, the Vital, Essential, Desire Analysis (VED Analysis) is employed. The primary objective is to measure how different medications vary in terms of their significance. This is achieved by assigning the letters V, E, and D, representing Vital (high importance), Essential (moderate importance), and Desire (low importance), respectively, to each medication category.⁴⁻⁸

In a study conducted by Nugraha⁹, a dataset comprising fast and slow-moving pharmaceutical products was utilized for clustering analysis. This approach leveraged the research of Ketkar and Vaidya¹⁰ who employed multiple categorization methods for comparative analysis. They identified the use of HML (High, Medium, Low), FSN (Fast, Slow, Non-movement), and SDE (Scarce, Difficult, Easy) categorizations. Each of these analyses served specific purposes:

- **HML Analysis:** It specifically assessed the value of each individual pharmaceutical item to categorize them according to their monetary worth.
- **FSN Analysis:** This analysis focused on the speed of movement of pharmaceutical products to distinguish between fast-moving and slow-moving items.
- **SDE Analysis:** It analyzed the likelihood of procurement or acquisition difficulties for pharmaceutical products.

The use of multiple attributes allowed decision-makers to gain a greater understanding of the unique characteristics of each pharmaceutical group, facilitating a broader perspective and enabling the application of developed data for forecasting

purposes. The researcher, drawing from various studies, chose to utilize these attributes to pre-group pharmaceutical items. These attributes included the annual cost value of medications to indicate their financial significance, the velocity of medication movement to represent their turnover speed, and the variability of medication objectives. These attributes were incorporated to demonstrate clinical importance, enabling the subsequent grouping of medications for more effective analysis.

Clustering algorithm

Clustering refers to a process of grouping data using unsupervised machine learning. In this approach, there is no pre-existing classification or guidance from a human expert. The tool used conducts its analysis and grouping of data autonomously. Unsupervised clustering differs from supervised classification, as it is often considered a more challenging task, given the absence of labelled data to serve as examples for the tool.¹¹ The objective of clustering is to uncover the genuine groups or patterns within a dataset, utilizing statistical principles for quantitative comparisons. It involves numerical data and the measurement of the numerical distances between data points, with closer data points indicating higher similarity. Clustering techniques can be broadly categorized into two major types: Hierarchical and Partitioning techniques.¹²

Partitioning clustering involves dividing data into a predetermined number (k) of clusters without establishing a hierarchical structure. This partitioning relies on specific criteria for grouping data points, with the Euclidean distance metric being a commonly utilized criterion as calculated by:

$$d(p, q) = \sqrt{\sum_{i=1}^n (q_i - p_i)^2}$$

where q_i and p_i denote the respective components along the "i" dimensions of the points q_i and p_i . Euclidean distance measures the proximity between data points, with closer points forming the same cluster. Various algorithms are employed for partitioning clustering, including k-Means, Partitioning Around Medoids (PAM), CLARA, CLARANS, Fuzzy C-Means, and DBSCAN. These algorithms play a crucial role in the creation

of distinct clusters and are particularly valuable for data analysis and pattern recognition tasks.

In Jain¹², k-means clustering has been the most widely used clustering approach since its introduction in the field of clustering half a century ago. It has been widely utilized in diverse study domains, including data mining, medical data analysis, customer segmentation, and image processing, showcasing its adaptability in fulfilling the goals of distinct users. Jain also emphasized other benefits of k-means clustering, such as its ease of use and understanding, efficient computing due to its direct process, and appropriateness for clustering vast amounts of data. In addition, there are pre-built software packages accessible, which eliminate the necessity for customers to create their own programs. Nevertheless, Jain also addressed certain constraints of k-Means clustering. A significant limitation is the requirement to predefine the number of clusters, which can provide difficulties for users without a full understanding of the data. In addition, k-means clustering is not suitable for analyzing complicated multidimensional data or data with significant amounts of outliers, as these characteristics might lead to instability in the cluster centroids, consequently impacting the accuracy of the clustering outcomes. Due to the simplicity, effectiveness, and widespread use of the k-means clustering algorithm, the researcher has adopted this technique for clustering drug demand.

K-Means clustering functions as demonstrated in Figure1:

- Step 1: The user specifies the number of clusters (k) into which the data should be divided.
- Step 2: Randomly select k initial data points as cluster centroids and assign data points to the cluster whose centroid they are closest to.
- Step 3: Once the initial clustering is established, compute the centroids of each cluster (k centroids).
- Step 4: Recalculate the cluster assignments for each data point by measuring the distance between the data point and the newly computed cluster centroids. If a data point is closer to the centroid of another cluster, it switches to that cluster.
- Step 5: Check if the cluster centroids have changed. If they have, repeat steps 3 and 4. Continue this process until the cluster centroids no longer change.

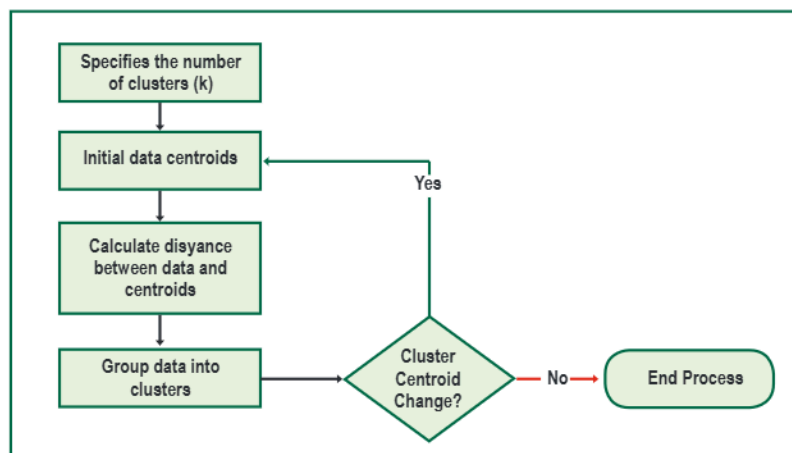


Figure 1: K-Means clustering process.

The k-means algorithm iteratively refines the cluster assignments and updates the centroids until a stable configuration is reached. This algorithm is appreciated for its simplicity and efficiency in partitioning data into distinct clusters based on proximity to cluster centroids.^{11,13}

In the investigation conducted by Nugraha⁹, an analysis was carried out on the categorization of pharmaceutical products through the application of chemical grouping methodologies. The study revealed that employing these techniques proved to be effective in clustering medications according to patient-specific requirements. The data gleaned from this research held significant utility in the context of future demand projection procedures. Medications from the preceding three years, in terms of their usage patterns, were sorted into distinct groups and subjected to strict control, particularly for those exhibiting high consumption rates and rapid turnover, in order to mitigate the risk of inventory shortages. Conversely, slower-moving medications were also carefully monitored to minimize errors in stock management. Hence, this study opted for the clustering method to categorize medications based on a multitude of interrelated factors, subsequently utilizing these categorized medications to identify resource allocation strategies with the least margin for error.⁹

Materials and Methods

Data Acquisition

1. Collecting data on pharmaceutical products monthly usage

During the data acquisition, the researcher submitted requests for data to the hospital under investigation. These data requests encompassed valuable information for each month regarding all 1,501 pharmaceutical products from January 2018 to December 2019, spanning a total duration of 24 months. The researcher employed a data retrieval table, which stipulated that the following data had to be provided:

- Cost data for each pharmaceutical unit.
- Monthly usage data for pharmaceutical products in the years 2018 and 2019.
- Medication movement data, as reflected by the average monthly utilization rates.
- A list of all medications to be considered within the hospital's inventory.

Once the usage data has been obtained, the data for each pharmaceutical product is utilized to calculate the average movement, as outlined in (2). This equation is employed to derive information regarding medication movement¹⁴:

$$\text{Average Movement} = \frac{\text{Total Usage}}{\text{Total Months}} \quad (2)$$

Where 'Total Usage' represents the cumulative usage of the pharmaceutical product. 'Total Months' is the total number of months under consideration. This calculation provides insights into the average movement of pharmaceutical products over the specified time period.

2. Collecting data on pharmaceutical products characteristics

This research classifies drugs into group based primarily on the ADE.^{4,7} Cost data for each pharmaceutical product is used in conjunction with usage quantity to calculate ADE as defined in (3).⁷

$$\text{ADE} = \text{Cost of the drug} \times \text{Quantity used/ year} \quad (3)$$

As for the data collected for VED Analysis, the researcher collected information through the input of all three individuals, namely physicians, pharmacists, or pharmacy inventory personnel. In cases where the data exhibited significant variability and could not yield a consensus, an additional medical staff member, specifically the Head of the Pharmacy Department, provided input. Data collection was not restricted to a specific timeframe; instead, the researcher collected data incrementally until the dataset was complete. The selected personnel provide values of V, E, or D based on their individual assessments. The researcher then set up these values in the VED Analysis Data table to determine the mode rankings. The specific definitions for each category of Analysis are provided to the staffs, as detailed below:

- **Vital (V):** These are medications that have a direct impact on a patient's life and treatment outcomes. In the event of a medication stockout reaching the patient (Stock out to patient), there is a high risk that the overall treatment could be compromised, potentially leading to life-threatening situations, temporary or permanent disability. Although substitute medications may be available, they may not have equivalent clinical effectiveness.
- **Essential (E):** Essential medications have a moderate level of importance and can influence overall treatment outcomes. They are crucial for patients who respond well to specific drugs. If patients do not receive these medications, it may affect the treatment approach. Essential medications can be substituted with other medications, but their clinical effectiveness may be somewhat similar.
- **Desire (D):** These medications have low significance and minimal impact on treatment outcomes. Whether patients receive these medications or not, they can be easily substituted with other drugs without significant impact on the patient's condition. They are often associated with simple diseases or minor ailments that do not pose a severe health risk.

Drug Clustering

1. Data Preparation

Before creating a cluster, data preparation processes are employed to transform the data into a suitable format. These processes were conducted using the RapidMiner program, with the steps outlined as follows:

- Step 1: Import the patient data and VED Analysis data.
- Step 2: Merge the two datasets into a single dataset.
- Step 3: Identify missing values and replace with '0'.
- Step 4: Calculate the average and variance of monthly demands.

- Step 5: Normalize clustering features using z-transformation method.

2.Data Clustering

After the data preparation process, k-means clustering was applied using the RapidMiner program. The key features utilized include annual drug expenditure, average movement, and demand variance. In this study, the appropriate number of clusters or 'k' value was determined by exploring a range from 2 to 20 clusters. The average within-centroid distance, which represents the average distance of each data point to the centroid of its cluster, was calculated. The suitable number of clusters was identified at the point where adding more clusters did not significantly decrease the average within-centroid distance. Once the optimal 'k' value was established, k-means clustering was implemented with the selected 'k'.

3.Cluster Selection

Further analysis was conducted to assess the distinctive characteristics of each cluster by comparing annual drug expenditure, average movement, and demand variance within each group. Recommendations were provided for each cluster.

Results

Data Collection and preparation

The drugs demand dataset encompasses the subsequent particulars for each medication type as Table 1 shows:

- Drug ID: A distinctive identifier for the drugs.
- Cost: The associated cost for the drug.
- Movement: Data concerning drugs mobility, which can exhibit variations among different drug types.

The characteristics of drugs have been processed by VED analysis as depicted in Table II. Based on the input and insights provided by a minimum of three healthcare experts or personnel within the hospital setting, an additional pharmacist will be included if consensus or conclusive findings cannot be reached.

Furthermore, the data collection yielded a total of 1,501 medication listings, categorized as follows: 213 listings in the "Vital" category (V), 588 listings in the "Essential" category (E), and 700 listings in the "Desire" category (D), constituting proportions of 14.19%, 39.17%, and 46.64%, respectively, in that order as depicted in Figure 2.

Table 1: Example of drug demand database

Drug ID	UOM	COST	1/2018	2/2018	3/2018	4/2018	5/2018	6/2018	7/2018	8/2018	9/2018	10/2018	11/2018	12/2018
WJN3ACXT7VCVQEG	Can	200.00	8	2	6	3	1	3	7	3	0	1	6	1
53GRQGQU99VGAM7	Piece	144.45	42	47	56	39	54	33	27	46	33	28	23	35
C9WZRGHAW85FHSC	Set	5.89	239	87	321	600	193	232	149	156	195	85	155	145
Q7YA76YE8GP5VEL	Bottle	12.94	1	20	6	9	6	3	5	9	8	3	3	4
BUG7DLHP2VDJQEE	Sachet	8.92	6	10	0	5	0	0	6	10	0	0	10	0
LFCUFCZH6E5XDAY	Tablet	0.65	449	335	297	211	461	209	390	248	158	752	303	462
RV7R6EFMDXBJX7D	Capsule	58.85	364	552	567	326	540	123	338	380	249	668	338	529
8T3MBLTGN5Q2MTS	Capsule	70.62	258	307	384	139	103	156	332	209	168	356	239	440
BXU7RZZVYRF68GC	Tablet	10.70	542	1025	929	822	1372	912	965	663	328	737	321	416
7X85GAXHK653GRE	Bottle	161.61	110	105	82	87	96	73	56	89	63	55	45	52
5ZT35BD2FQ6SX2E	Sachet	16.81	17	20	51	9	39	10	6	42	51	181	262	511
VM75BRAAXRZEHMV	Tablet	32.10	7	24	15	12	10	118	47	78	91	27	43	60
9MU3Y72P5C5X6A4	Set	321.00	39	41	58	32	30	37	38	61	43	20	32	36

Table 2: Example of vital, essential, desire (VED) Analysis results

Drug ID	Staff #1	Staff #2	Staff #3	Staff #4	Conclude
JEEKH826ZZ37QBZ	D	V	E	D	D
JMEYPQRDA746XJM	V	D	V		V
XRFXZV56LFEKP3U	E	E	E		E
79QJAAACHEVBJQN9	V	D	V		V
K7MGLZJA2Z959GN	D	D	E		D
CZDGD4H4EWEAGVW	D	V	E	D	D
BZ9DJPEYFYRAMAL	E	V	E		E
TWQ6YXBVDM9RECU	V	E	V		V
RWEEBSMCNUZPQXB	V	D	V		V
WA5T3N587P6NCJU	V	V	E		V

The dataset from Table 1 and Table 2 will be aggregated into a primary dataset by RapidMiner program for the purpose of subsequent clustering. The data preparation process is depicted in Figure 3. The data preparation process involves using multiple program operators that are connected to one another, creating a sequential flow in the process. As shown in Figure 3, two datasets are joined, missing values are cleaned, average movement per month is calculated, and demand variance is generated. This interconnected sequence of operations ensures that the data is appropriately processed and refined for analysis. The example of generated calculation from the program is shown in Table 3.

In this study, the focus will be on clustering the “Essential; E” category of drugs, as the “Vital; V” and “Desire; D” categories can be managed using straightforward inventory management principles and established guidelines. This matter will be further elaborated upon in the Discussion section.

Data Clustering Results

The RapidMiner process used to determine the ‘k’ value is shown in Figure 4. The results of the ‘k’ value determination are presented in Figure 5. The appropriate value of ‘k’ is determined based on the ‘elbow’ point of the curve, where the slope of the curve changes. In this study, the appropriate value for ‘k’ is determined to be 6.

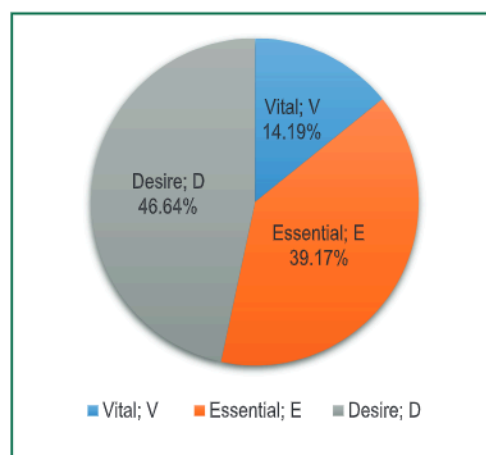


Figure 2: shows vital, essential, desire (VED) analysis results.

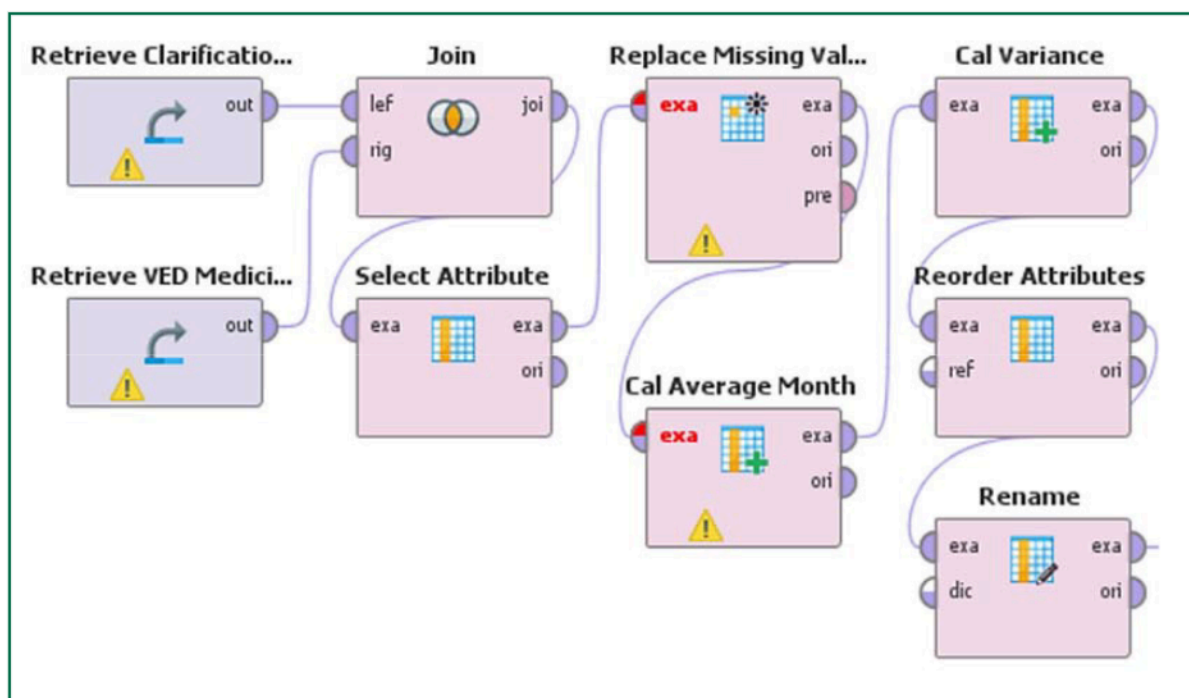
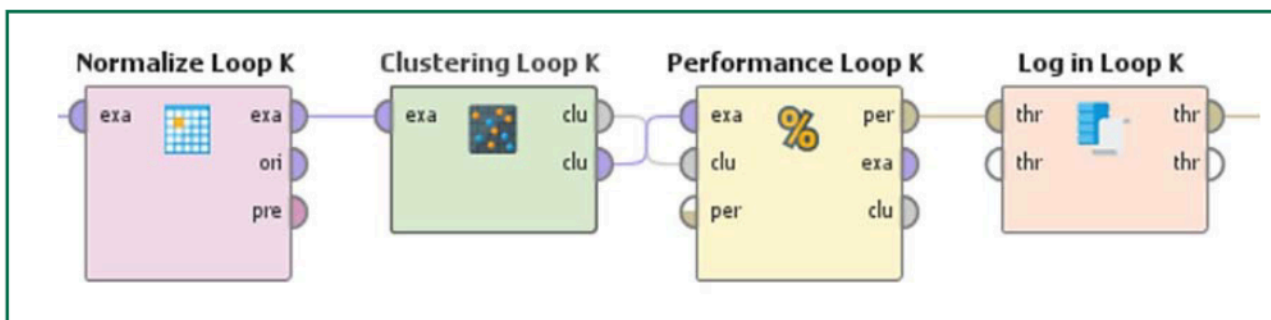
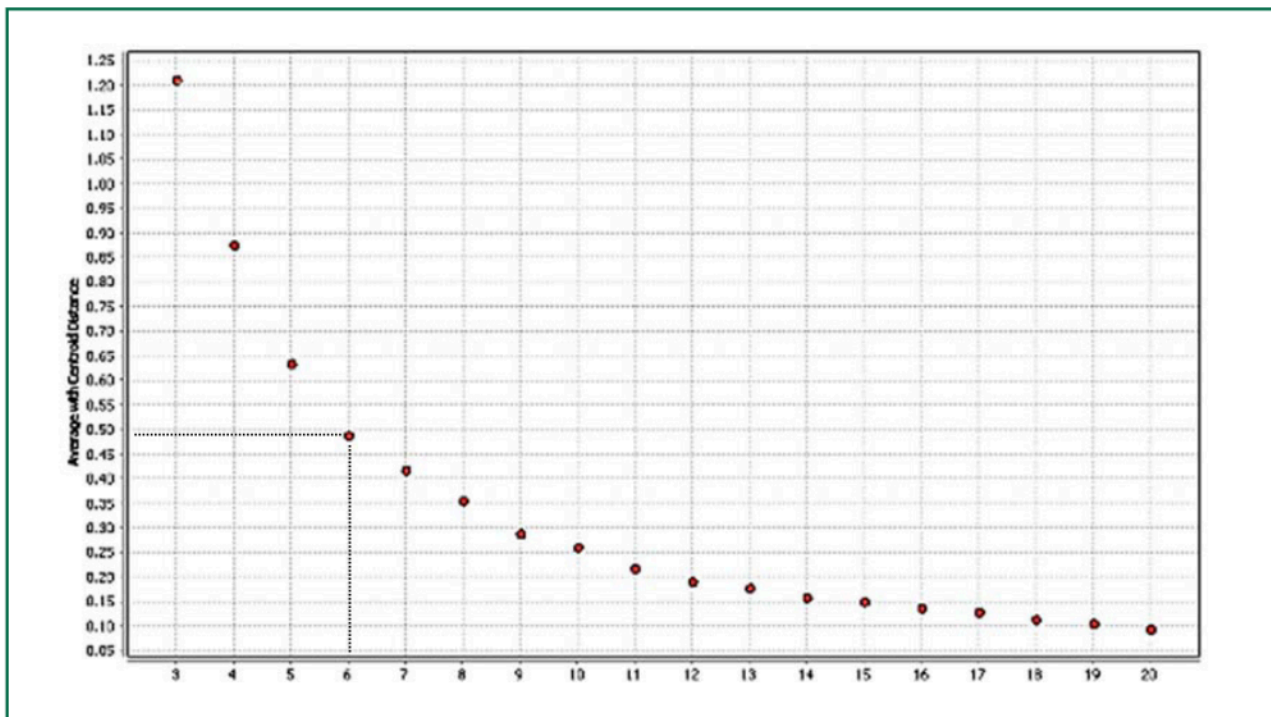


Figure 3: RapidMiner process used for data preparation.

Table 3: Example of Calculation Result

Drug ID	ADE (Annual Drug Expenditure)	Average Per Month	Variance
CL2T7XX6SE2G9XY	2,640,917.825	309.292	1,870.129
FZ2EX743LKQ693J	2,293,898.100	517.833	14,621.536
FAQJCDQK28GZHYC	2,267,195.505	91.333	186.580
DH4TJF48TTQ92LN	1,714,059.824	75.708	422.824
UDREN32EZJUYJTY	1,352,557.803	1,997.792	946,913.911
DJP8BSBZT8SJB3J	1,315,679.076	2,961.208	2,042,304.955
VZ83B2H7ZS68627	1,206,732.141	174.042	1,124.911
W4AHLM78DN48FVV	1,182,885.000	33.500	33.652

**Figure 4:** RapidMiner process used to determine 'k'.**Figure 5:** A chart between number of cluster and average within centroid distance

The *k*-means clustering model was constructed using the RapidMiner program. The process is visually represented in Figure 6. The procedure encompassed the stages of data normalization, clustering, and the extraction of cluster prototypes.

The clustering results are presented in Table 4 and Figure 7, which facilitates a comparison between the six clusters. Table IV presents the characteristics of cluster_0 to cluster_5, with calculated values for each cluster, such as ADE, average per month, and variance. Figure 7 displays a scatter plot of these clusters. In the scatter plot, each data point is represented based on three dimensions: the x-axis corresponds to average movement per month, the y-axis represents ADE, and the z-axis depicts variance. This visualization provides a complete view of the distribution and relationships within the clusters.

Table 4 clearly shows that Cluster_0 (69 items) has a modest annual drug expenditure, along with moderate levels of drug mobility and variance. Cluster_1 (17 items) exhibits a comparatively low yearly drug expenditure, although it is distinguished by a high level of drug movement and variability. Cluster_2 (454 items) exhibits the most minimal values for all parameters, encompassing annual drug cost, monthly mobility, and variance. On the other hand, cluster_3 (41 items) has the second largest yearly drug spending out of the six clusters,

along with comparatively little mobility and variability. Cluster_4 (4 items) is characterized by its medication expenditure, which is about equal to that of Cluster_3. However, it exhibits significant mobility and variability. Cluster_5 (3 items) has the largest annual drug spend, although it has the lowest amounts of movement and variance.

Discussion

This study used a combination of VED (Vital, Essential, Desirable) analysis and k-means clustering to categorize essential medications. The primary goal was to thoroughly classify medications, distinguishing between their distinctive qualities, and identifying significant medication groups.

By taking into account the opinions of doctors, pharmacists, and inventory staff, the VED classification method accurately figures out the significance drugs are for treating and preventing diseases. Three categories were created out of the 1,501 drugs in the study: Essential (588 items), Desirable (700 items), and Vital (213 items). According to the study, controlling the demand for both necessary and desirable drugs is actually rather simple, as the discussion section illustrates. Essential drugs are more complex because they only partially contribute to therapy and have a variable effect on patient outcomes.

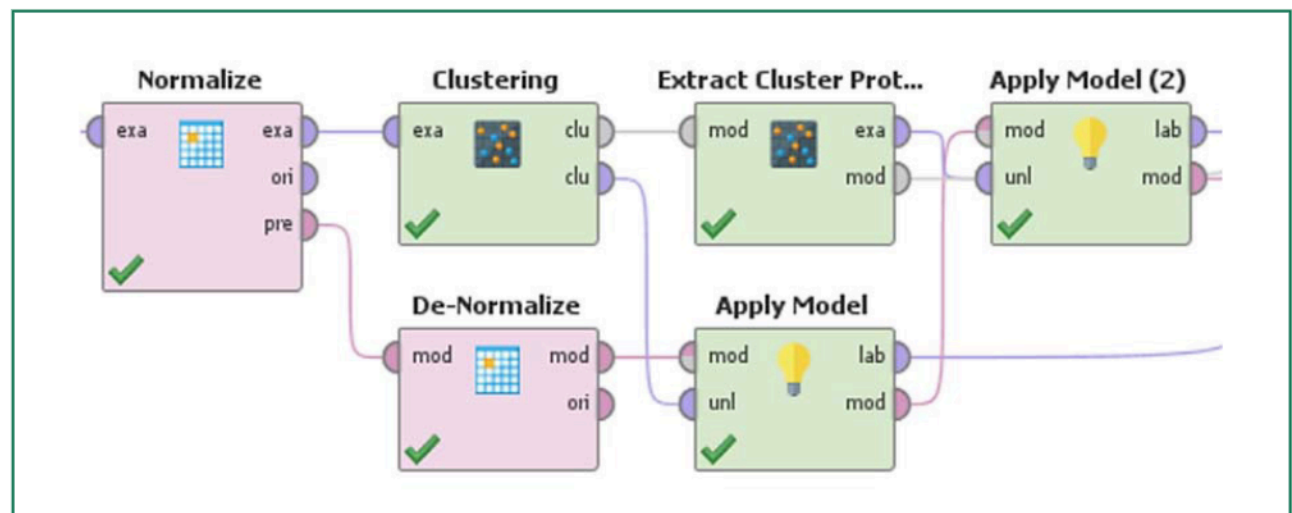


Figure 6: Clustering using the k-Means clustering technique in the RapidMiner.

Table 4: Clustering Value Results.

Row No.	cluster	ADE (Annual Drug Expenditure)	Average Per Month	Variance
1	cluster_0	92,123.139	617.768	617.768
2	cluster_1	158,637.894	1,609.039	1,609.039
3	cluster_2	33,086.996	60.866	60.866
4	cluster_3	396,009.557	298.946	298.946
5	cluster_4	365,098.071	3,390.885	3,390.885
6	cluster_5	1,730,280.975	209.014	209.014

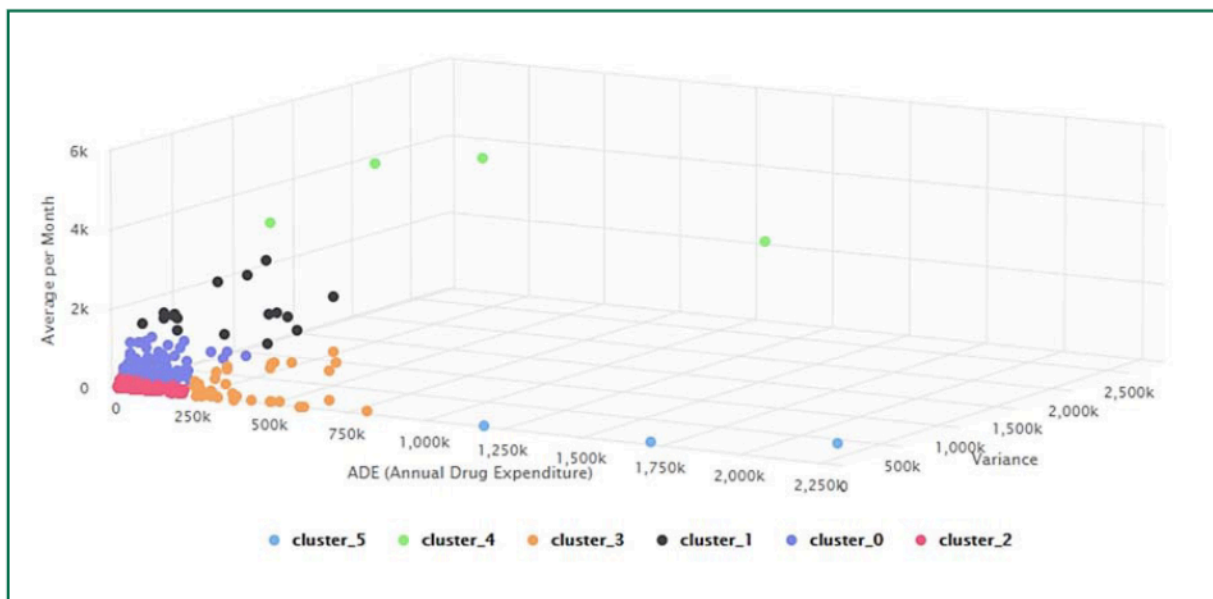


Figure 7: Three-dimensions scatter plot of the k-Means clustering result.

As a result, the researcher used k-means clustering to do research on essential drugs. This necessitated analyzing variables including variation, average monthly demand, and annual drug cost. The clustering results are very important because they offer detailed information about each group, which makes it possible to implement more precise and focused treatment plans for various drugs.

The VED Analysis conducted prior to clustering revealed that drugs in the “Essential” (E) category are more suitable for clustering to determine appropriate management strategies compared to those in the “Vital” (V) and “Desirable” (D) categories for the following reasons:

For the “Vital; V” drug class, which bears significant implications for patient welfare, a high service level is imperative. This entails adopting a zero-tolerance stance towards stockouts to ensure an uninterrupted supply of essential medications to patients. Consequently, medications falling within the “Vital; V” category require continuous reservation and the establishment of judicious safety stock levels. This proactive approach is indispensable for addressing exigencies that may arise and for preserving the well-being of patients. The insistence on maintaining a safety stock for these medications underscores its critical importance within the hospital setting.

For medications classified under the “Desire; D” category, deemed to have a minimal impact on treatment, this aligns with the hospital’s policy of assigning lower priority to this drug group. The management approach involves simple methods, such as calculating straightforward averages by experienced personnel or adhering to minimum order requirements specified by manufacturers.

For the clustering results obtained through the k-means method, the researcher found that each group differs in terms of annual drug expenditure, monthly movement, and variance. The researcher has analyzed the key characteristics and management strategies for each drug group as follows:

- Cluster_0: When compared to other clusters, Cluster_0 shows relatively low annual drug expenditure, but it ranks third out of six in terms of average monthly movement and variance. According to López, Mendoza 15, managing unstable demand can be achieved by increasing the reorder point to maintain sufficient safety stock. This increase in the reorder point helps ensure enough safety stock to accommodate demand variability. Additionally, since the annual expenditure of these drugs is low, increasing the safety stock has a minimal impact on the overall inventory value.
- Cluster_1: This cluster also demonstrates low annual drug expenditure, with moderate average monthly movement and the second-highest variance among the six clusters. This group can be managed without complex forecasting methods, using the same inventory management strategies as Cluster_0. This includes calculating the reorder point based on demand and possibly implementing a continuous review model or a periodic review model. These approaches help manage high-variance, low-value drugs effectively.
- Cluster_2: This cluster has the lowest values for both annual drug expenditure and average monthly movement, as well as very low variance. These characteristics indicate that managing Cluster_2 requires minimal effort within the hospital since the monthly demand for these drugs is low. Therefore, this cluster can be efficiently managed by setting an appropriate reorder point and ordering drugs as needed.

- Cluster_3: Exhibits relatively high annual drug expenditure compared to other clusters. This characteristic makes it critical to accurately forecast future medication needs to ensure adequate inventory management without overstocking. And despite high annual drug expenditure, Cluster_3 shows low variability and dispersion in monthly medication usage. This stability allows for more predictable forecasting models, reducing the risk of stockouts or excess inventory. Forecasting models such as time series methods or machine learning can be effectively applied.¹⁶⁻¹⁸
- Cluster_4: The hospital should prioritize forecasting for this cluster due to its high annual expenditure, which significantly impacts the overall inventory value. Additionally, the high variance and turnover make it challenging to forecast demand based solely on staff experience, increasing the risk of errors in order estimation. Due to the variability in monthly usage, traditional forecasting methods might not suffice for Cluster_4. Advanced techniques such as machine learning algorithms (e.g., Random Forest, Neural Networks) or hybrid models that combine statistical methods with machine learning can provide more accurate forecasts.^{17,19}
- Cluster_5: This cluster has the highest annual drug expenditure but the lowest average monthly movement and variance. The high annual expenditure combined with low average monthly usage indicates that these drugs are specialized for specific cases or conditions, affecting a small patient group. To address this, the hospital should establish policies and agreements involving all relevant stakeholders, including physicians, pharmacists, and patients. Setting specific treatment dates and creating an appointment system will allow pharmacists or purchasing staff to order and dispense these drugs according to a scheduled timeline.

After conducting the study, the researcher worked with hospital staff to put the experimental findings into practice. The hospital team had a positive response to the idea of medication clustering using machine learning and indicated a desire to customize it for their situation. As a result, the researcher set up the program such that it might be used in the future by the pharmacy team, inventory personnel and business analyst. Nevertheless, this study did not include a satisfaction survey of hospital related staff who had left to see how they responded to the results. Instead, the researcher and the hospital team worked together closely from the start to the finish to come up with the clustering method.

References

1. Bijvank M, Vis I. Inventory control for point-of-use locations in hospitals. *JORS*. 2012;63:497-510. doi:10.1057/jors.2011.52.
2. Kelle P, Woosley J, Schneider H. Pharmaceutical supply chain specifics and inventory solutions for a hospital case. *ORHC*. 2012;1(2):54-63. doi: 10.1016/j.orhc.2012.07.001
3. Zwaida TA, Pham C, Beauregard Y. Optimization of inventory management to prevent drug shortages in the hospital supply chain. *Appl Sci* 2021;11:2726. doi: 10.3390/app11062726.
4. Anand T, Ingle GK, Kishore J, et al. ABC-VED analysis of a drug store in the Department of Community Medicine of a Medical College in Delhi. *Indian J Pharm Sci*. 2013;75(1):113-7. doi: 10.4103/0250-474X.113543.

Conclusion

This study specifically focuses on facets of drug properties. Future study could broaden its scope by incorporating more targeted assessments within pharmacological medication classes, considering clinical characteristics, and expanding investigations into the likelihood of shortages of drugs with limited expiration dates. Adopting this more expansive approach would help our understanding of pharmaceutical inventory management and offer better insights into dealing difficulties related to certain drug groups and operational limitations. The findings of this study's clustering are similar to those of Nugraha⁹ research, which demonstrated that clustering effectively groups drugs with similar characteristics. These groups help identify trends such as high demand, seasonal variance, and critical stock levels, leading to better inventory management. Finally, this improves procurement plan, reduces costs, reduced losses from expired drugs, and emphasize decision-making in planning, tracking and inventory control.

In this research, VED analysis and k-means clustering are successfully cluster drugs into groups. The potential for customized utilization to satisfy specific needs is demonstrated by the collaboration between a researcher and hospital staffs in doing these studies. Further studies may build on these study result and increase pharmaceutical inventory control by including more attributes such as considering drug scarcity or pharmacological properties. The use of clustering techniques in demand forecasting and inventory control enhancement shows how effective k-means is in optimizing hospital drug inventory. This could lead to improved procurement planning, cost savings, and improved decision-making. Hospital staff responsible for drug inventory management acquire fresh perspectives and guidelines for inventory policies, which can be effectively implemented as situation evolve. In addition to being able to spend more time to document and quality tasks, they can use the time to prioritize the clusters from the analysis and ultimately cut down on work hours in the future. Newer machine learning technologies, like deep learning or machine learning models, might be considered to accurately predict how people will use pharmaceuticals in the future. Using cluster data will make it simple for these models to correctly forecast and make understanding of huge amounts of data.

5. Fitriana I, Satria RGD, Setrawan DCB. Medicine Inventory Management by ABC-VED Analysis in the Pharmacy Store of Veterinary Hospital, Yogyakarta, Indonesia. *Asian J Anim Vet Adv*. 2017;13(1):85-90. doi: 10.3923/ajava.2018.85.90
6. Hussain M, Siddharth V, Arya S. ABC, VED and lead time analysis in the surgical store of a public sector tertiary care hospital in Delhi. *Indian J Public Health*. 2019;63(3):194-8. doi: 10.4103/ijph.IJPH_282_18.
7. Kumar S, Chakravarty A. ABC-VED analysis of expendable medical stores at a tertiary care hospital. *Med J Armed Forces India*. 2015;71(1):24-7. doi: 10.1016/j.mjafi.2014.07.002.
8. Lestari F, Ulfah A, Azwar B, et al. Combining ABC and VED analysis for managing medicine inventory (case study at community development elderly in Indonesia). *Int J Adv Scie Engineer Informa Technol*. 2019;9(3):952-9.
9. Nugraha JAM. Medicine Inventory Grouping using Clustering Data Mining. *Indonesian J Inform Syst*. 2019;2:33-44. doi: 10.24002/ijis.v2i1.2340
10. Ketkar M, Vaidya OS. Developing Ordering Policy based on Multiple Inventory Classification Schemes. *Proc Social Behav Scie*. 2014;133:180-8. doi: 10.1016/j.sbspro.2014.04.183.
11. Saxena A, Prasad M, Gupta A, et al. A review of clustering techniques and developments. *Neurocomputing*. 2017;267:664-81. doi: 10.1016/j.neucom.2017.06.053
12. Jain AK. Data clustering: 50 years beyond K-means. *PaReL*. 2010;31(8):651-66. doi:10.1016/j.patrec.2009.09.011
13. Likas A, Vlassis N, Verbeek JJ. The global k-means clustering algorithm. *Pattern Recognition*. 2003;36(2):451-61. doi:10.1016/S0031-3203(02)00060-2.
14. Axsäter S. Inventory Control: Thrid edition. 2015;225. ISBN: 978-3-319-157283
15. López JA, Mendoza A, Masini J. A classic and effective approach to inventory management. *International J Indust Engineer Theor Appli Pract*. 2013;20:372-86.
16. Fattah J, Ezzine L, Aman Z, Moussami HE, et al. Forecasting of demand using ARIMA model. *Int J Engineer Busine Manage*. 2018;10:184797901880867. doi:10.1177/1847979018808673.
17. Tarallo E, Akabane GK, Shimabukuro CI, et al. Machine learning in predicting demand for fast-moving consumer goods: an exploratory research. *IFAC-PapersOnLine*. 2019;52(13):737-42. doi: 10.1016/j.ifacol.2019.11.203
18. Merkurieva G, Valberga A, Smirnov A. Demand forecasting in pharmaceutical supply chains: a case study. *Procedia Comp Scie*. 2019;149:3-10. doi:10.1016/j.procs.2019.01.100
19. Ray A, Chaudhuri AK. Smart healthcare disease diagnosis and patient management: Innovation, improvement and skill development. *Machine Learn Applic*. 2021;3:100011. doi: 10.1016/j.mlwa.2020.100011